Toward Principled Threshold Selection for Blink Region Detection in Biological Signals: A Cross-Dataset Benchmark with LLM-Guided Candidate Derivation

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Abstract

Blink detection from electroencephalography (EEG) serves two complementary purposes: blinks are a major physiological contaminant that must be identified for artifact handling, and they are also informative behavioral markers linked to arousal, fatigue, cognitive load, and operational performance. Despite substantial progress in blink detection, a practically important setting remains comparatively under-addressed: epoch-structured recordings in which a long continuous signal is divided into non-overlapping analysis windows. This setting is the norm in driving neuroscience and other neuroergonomic paradigms, yet existing detectors—including BLINKER and MNE-based amplitude routines—operate on concatenated signals and implicitly pool blink-containing and blink-free epochs when estimating thresholds. We show that this pooling may bias threshold placement and alter the precision-recall balance. We propose a three-stage epoch-aware pipeline. Stage A screens epochs by their peak-to-peak amplitude via an autoreject-inspired cross-validation criterion, yielding a set of suspicious (likely blink-containing) epochs. Stage B estimates a robust sample-level threshold from the suspicious epoch pool using the median and median absolute deviation (MAD), concentrating estimation on signal segments enriched for blink-like excursions. Stage C applies the threshold to extract blink regions from the suspicious epochs and maps detections back to epoch-local coordinates. A fallback mechanism ensures stability when too few suspicious epochs are available. We evaluate four conditions—BLINKER-concat, MNE-annot, Proposed-Mean, and Proposed-Med—on two naturalistic driving-EEG corpora, Raja and Cao2018, under epoch durations of 30, 40, and 60 seconds, with 30 seconds selected as the primary setting. Statistical comparisons use macro-averaged event-level F_1 with Wilcoxon signed-rank tests. Additional analyses examine stability across boundary tolerances. Results demonstrate that epoch-aware threshold estimation consistently outperforms naive concatenation, that the robust MAD-based estimator is preferable to a mean-based alternative, and that the robust median configuration (Proposed-Med) achieved the best macro-averaged event-level F_1 (pooled 0.867) and significantly outperformed the baselines and the mean-based variant (Wilcoxon, Bonferroni).

1 Introduction

Blinking is a physiological process that involves the closure and reopening of the eyelids(Nyström et al., 2024). Its primary functions are to protect the ocular surface, maintain corneal moisture, and support visual comfort(Nyström et al., 2024). Beyond these physiological functions, blink behaviour has been increasingly recognised as an important indicator in fatigue-driving research(Alyan, 2023; Tian and J. Cao, 2021). Variations in blink-related features, such as blink frequency, blink duration, eye-closure duration, and blink patterns, can reflect changes in alertness and cognitive state(Alyan, 2023; Ren et al., 2019). As fatigue increases, drivers may exhibit slower or prolonged blinks, which can indicate reduced vigilance and a higher risk of impaired driving performance (Alyan, 2023). Therefore, blink behaviour provides a useful and non-invasive marker for monitoring driver fatigue(Shahbakhti et al., 2023; Tian and J. Cao, 2021).

In the neuroscience study of fatigue driving, blink activity can be captured through external methods, such as camera-based eye monitoring, or through electrophysiological recordings, particularly electroencephalography and electrooculography (Tian and J. Cao, 2021; Shahbakhti et al., 2023). Although EEG is mainly used to measure brain activity, blink-related components are often present in EEG signals because eyelid movements and eye movements generate strong electrical potentials (Croft and Barry, 2000; Akuthota et al., 2023). These ocular activities are especially prominent in frontal EEG electrodes due to their close proximity to the eyes (Croft and Barry, 2000; Kleifges et al., 2017). As a result, blink information can be extracted from EEG recordings, either by analysing the ocular components embedded in the EEG signal or by combining EEG with EOG-based measurements (Agarwal and Sivakumar, 2019; Shahbakhti et al., 2023). This makes EEG/EOG recordings valuable not only for studying neural activity during fatigue but also for identifying blink-related markers associated with drowsiness (Ko et al., 2020; Alyan, 2023).

It is important to distinguish between two major perspectives on blink activity in the existing literature (Croft and Barry, 2000; Tian and J. Cao, 2021). In many EEG studies, blink-related activity is treated as an artifact because it contaminates neural signals and may interfere with the interpretation of brain activity (Croft and Barry, 2000; Akuthota et al., 2023). In such cases, blink components are commonly removed during EEG preprocessing (Croft and Barry, 2000; Klein and Skrandies, 2013). However, in fatigue-driving studies, blink activity can also be treated as meaningful information rather than noise (Tian and J. Cao, 2021; Shahbakhti et al., 2023). From this perspective, blink events are identified, extracted, and analysed as potential indicators of driver fatigue (Ko et al., 2020; Shahbakhti et al., 2023). Blink detection methods can generally be classified into machine-learning-based, morphology-based, and thresholding-based approaches (Agarwal and Sivakumar, 2019; Zhang et al., 2023). Machine learning methods rely on trained models to classify blink events, while morphology-based methods identify blinks based on their characteristic waveform shape (Rashida and Habib, 2023; Jiang et al., 2023). Thresholding-based methods, in contrast, detect blinks by comparing signal amplitudes with predefined or adaptive threshold values (Chang et al., 2016; Tran et al., 2021; Zhang et al., 2023).

Thresholding is one of the most straightforward and widely used approaches for detecting blink activity from EEG or EOG signals (Tran et al., 2021; Chang et al., 2016). In this method, candidate blink events are identified when the signal amplitude exceeds a certain threshold value, as blink activity typically produces large-amplitude deflections, especially in frontal channels (Chang et al., 2016; Kleifges et al., 2017). The simplicity of thresholding makes it computationally efficient and easy to implement, particularly in real-time fatigue-detection systems (Tran et al., 2021; Jefri et al., 2022). In addition, thresholding is often used as an initial detection step in more complex pipelines, where the detected candidates are further refined using morphological rules or machine-learning classifiers (Jiang et al., 2023; Ghosh et al., 2023). However, selecting an appropriate threshold value is not trivial (Guttmann-Flury et al., 2019; M. Wang et al., 2022). Blink amplitude can vary across participants, electrode locations, recording devices, signal quality, and experimental conditions (Guttmann-Flury et al., 2019; M. Wang et al., 2022). A threshold that is too low may produce false detections due to noise or other artifacts, whereas a threshold that is too high may miss genuine blink events (Tran et al., 2021; Jefri et al., 2022). Therefore, threshold selection remains a critical consideration in the reliable detection of blink activity for fatigue-driving research (Ko et al., 2020; Shahbakhti et al., 2023).

In many fatigue-driving studies, EEG recordings are segmented into epochs that correspond to specific task events or analysis windows (Z. Cao et al., 2019). This epoch-structured setting introduces additional challenges for threshold-based blink detection (Tran et al., 2021; Jefri et al., 2022). When thresholds are estimated from the entire recording, they may be biased by the presence of clean epochs that contain few or no blinks, which may distort threshold estimates and suppress blink detection in epochs where blinks are more prevalent (Klein and Skrandies, 2013; Guttmann-Flury et al., 2019). As a result, naive application of thresholding in an epoch-structured context may risk under-detection of blinks, which undermines the reliability of fatigue assessments based on blink metrics (Ko et al., 2020; Shahbakhti et al., 2023). Addressing this challenge requires developing blink detection methods that explicitly account for the epoch structure and can adapt threshold estimation to the specific characteristics of each epoch or subset of epochs, rather than relying on a global threshold derived from the entire recording (Guttmann-Flury et al., 2019; Agarwal and Sivakumar, 2019).

Thresholding remains a foundational stage in eye-blink detection because it is widely used both as a first line of defence and as a candidate-generation or preprocessing step that precedes more complex machine-learning-based and morphology-based pipelines (Chang et al., 2016; Kleifges et al., 2017; Tran et al., 2021; Zhang et al., 2023; Agarwal and Sivakumar, 2019). Some pipelines use thresholding as a candidate-generation step that precedes later refinement, refining its candidates with later stages rather than treating it as the final detector; however, this makes the thresholding stage itself worthy of focused study because its quality propagates downstream. This study therefore deliberately examines thresholding rather than claiming that thresholding is superior to machine learning: it is the simplest and computationally lightweight detector, making it suitable for real-time fatigue monitoring; it is frequently the first stage whose output is consumed by downstream refinement, so the reliability of thresholding bounds the performance of the whole pipeline; and the epoch-structured estimation bias studied here affects the threshold-estimation step specifically, meaning that improving this step can benefit any pipeline that begins with thresholding, regardless of the downstream method.

This paper makes several contributions to the field of blink detection in fatigue-driving research.(Alyan, 2023; Shahbakhti et al., 2023) First, we formalise the problem of blink detection in the epoch-structured setting and demonstrate the limitations of naive thresholding approaches that do not account for the epoch structure.(Tran et al., 2021; Jefri et al., 2022) Second, we propose a three-stage pipeline for blink detection that is, to our knowledge, among the first to include epoch screening, sample-level threshold estimation, and blink-region extraction, which explicitly handles the challenges of the epoch-structured setting. Third, we introduce a robust threshold estimator based on the median and median absolute deviation (MAD) and systematically compare it against a mean-based alternative (Proposed-Mean) and conventional baselines, including BLINKER-concat and MNE-annot. Finally, we provide a comprehensive benchmark of competing methods across 30, 40, and 60 s epoch durations and a boundary-tolerance sensitivity analysis, all evaluated on two naturalistic driving-EEG corpora, Raja and Cao2018, under an event-level overlap criterion with macro-averaged F_1 and Wilcoxon signed-rank tests.(Z. Cao et al., 2019; Dari et al., 2020)

2 Related Work

The classical literature established the difficulty of ocular contamination and motivated automated statistical detection directly from EEG. Early work emphasized artifact removal and statistical screening rather than blink-region benchmarking per se, but it already showed that extreme-value behavior, channel context, and morphology matter (Croft and Barry, 2000; Klein and Skrandies, 2013). This historical backdrop explains why simple one-value global thresholds can be attractive, yet brittle.

A first important stream comprises single-channel or few-channel threshold-centered detectors.

Chang et al. (2016) detect blink artifacts from a single prefrontal channel using digital filtering and rule-based decisions. Tran et al. (2021) present an explicitly thresholding-based EEG blink detector and compute a minimum amplitude threshold from transformed signal magnitude and signal standard deviation. Zhang et al. (2023) move toward real-time frontal EEG detection with short windows and decision logic around potential blink boundaries. M. Wang et al. (2022) extend thresholding to harder clinical settings by optimizing multidimensional features in the presence of epileptiform discharges, and later work based on higher-order cumulants continues the move toward short-segment single-channel detection (G. Wang, 2025).

A second stream contains hybrid thresholding pipelines. BLINKER first identifies candidate intervals with a mean-based standard-deviation threshold and then rejects implausible candidates using robust amplitude-distribution criteria derived from the median absolute deviation (Kleifges et al., 2017). J. Cao et al. (2021) use cascaded thresholding over amplitude, amplitude displacement, and cross-channel correlation before applying a Gaussian mixture model. Agarwal and Sivakumar (2019) self-learn user-specific profiles in an unsupervised manner, while Valderrama et al. (2018) use iterative template matching and automatic threshold/template estimation for suppression in single-channel recordings. Guttmann-Flury et al. (2019) further emphasize adaptation to inter- and intra-subject variability, underscoring that fixed thresholds are often not enough.

A growing stream of blink and ocular-artifact research replaces hand-tuned thresholds with data-driven models. Convolutional neural networks have been used both to detect eye-blink artifacts and to remove them from EEG (Jurczak et al., 2022; Gawande and Badotra, 2022), while transformer- and video-based architectures have pushed blink-detection accuracy further (Fodor et al., 2023; Hong et al., 2024). Autoencoder and segmentation-denoising networks also target single-channel artifact removal (Faisal et al., 2025; Li et al., 2023), and recent reviews catalogue the broader shift from conventional thresholding toward deep learning for physiological-artifact handling, including in wearable EEG (Akshath Raj et al., 2026; Arpaia et al., 2025). In driving and fatigue monitoring, machine-learning and CNN models detect drowsiness from blink and eye-state cues (Cabot et al., 2024; Makhmudov et al., 2024). These data-driven detectors can be accurate, but they typically require labelled training data, more computation, and offer less transparency than a single interpretable threshold. This contrast motivates studying the lightweight thresholding stage itself, which frequently precedes or complements such pipelines, and in particular the epoch-structure estimation bias that this paper addresses.

3 Method

3.1 Problem Definition

Let a blink annotation be an interval $g_i = [s_i, e_i]$ and a detector output be an interval $d_j = [\hat{s}_j, \hat{e}_j]$. A detected region is counted as a true positive if it overlaps at least one unmatched ground-truth region. Unmatched detections are false positives; unmatched ground-truth regions are false negatives. This event-level treatment follows the logic of interval-based detection rather than pointwise accuracy, which is appropriate because blink boundaries are uncertain, annotation granularity varies, and non-event samples dominate the timeline (Saito and Rehmsmeier, 2015; Salles, 2024).

3.2 Datasets

Two naturalistic driving-EEG corpora were evaluated: Raja and Cao2018. The Raja corpus comprised 46 complete sessions recorded with a 128-channel EGI system (ANT Neuro), referenced to CPz under a modified international 10-20 layout, during a simulated lane-keeping driving task;

ground-truth blink annotations were derived from human EAR/EOG labelling. The Cao2018 corpus was the sustained-attention driving corpus of Z. Cao et al. (2019), in which 32-channel EEG was recorded during an approximately 90-minute lane-keeping task; 58 complete sessions were used, and 4 non-complete sessions were excluded. Because the published preprocessed version of Cao2018 removed eye blinks, the raw recordings were used and paired with external blink-region annotations.

3.3 Preprocessing

All detectors used identical preprocessing, consisting of a 1–20 Hz band-pass filter, no resampling, and EEG-channel selection. The channel scope was dataset specific: for Raja, the analysis was restricted to a minimal frontal montage comprising three channels (E3, E9, and E22), whereas for Cao2018, all 32 EEG channels were provided to the pipeline and the best channel was selected separately for each session. Thus, Raja channel selection was confined to frontal sites by construction, while Cao2018 channel selection was performed over the full montage.

3.4 Epoch Segmentation and Detection Pipeline

Recordings were segmented into fixed-length epochs, with 30 s used as the primary epoch length and 10, 20, 40, and 60 s additionally evaluated in an epoch-sensitivity sweep. The proposed detector was implemented as a three-stage epoch-aware threshold pipeline. First, suspicious/bad epochs were screened using an autoreject-based procedure with a fixed random state and a requirement that at least one epoch be flagged. Second, a sample-level amplitude threshold was estimated from the screened epochs using a robust centre and a standard-deviation multiplier of 3.5. Third, threshold crossings were expanded into blink regions. The detailed equations for each stage were given in the following subsections rather than duplicated in this overview.

3.5 Selection of Bad Epochs

The first stage of the proposed pipeline performs epoch-level screening using an autoreject-inspired threshold selection procedure. Its purpose is not to estimate the onset or offset of a blink, but to identify epochs whose amplitudes are sufficiently abnormal to warrant subsequent sample-level localization. Let $X \in \mathbb{R}^{N \times C \times T}$ denote the epoched multichannel signal after preprocessing, where N is the number of valid epochs, C is the number of channels, and T is the number of samples per epoch. The sample at epoch $i \in \mathcal{E} = \{1, \ldots, N\}$, channel $c \in \mathcal{C} = \{1, \ldots, C\}$, and within-epoch time index $t \in \{1, \ldots, T\}$ is denoted by $x_{i,c,t}$.

For each epoch and channel, an amplitude-range statistic is computed using the peak-to-peak value

$$a_{i,c} = \max_{1 \le t \le T} x_{i,c,t} - \min_{1 \le t \le T} x_{i,c,t}.$$

Large values of $a_{i,c}$ indicate that the corresponding epoch contains a high-amplitude excursion in channel c, a pattern consistent with ocular or other transient artifacts. Rather than imposing a fixed rejection level, the pipeline estimates a channel-specific threshold $\tau_c^\star$ from the data. Candidate thresholds are taken from the empirical range of peak-to-peak amplitudes for that channel, denoted $\mathcal{Q}_c = \{a_{i,c} : i \in \mathcal{E}\}$, and are evaluated by a cross-validation criterion.

Specifically, let $\{(\mathcal{T}_k, \mathcal{V}_k)\}_{k=1}^{K}$ be cross-validation splits of the epoch indices into training sets $\mathcal{T}_k$ and validation sets $\mathcal{V}_k$. For a candidate threshold $\tau \in \mathcal{Q}_c$, the training epochs retained for channel c in fold k are

$$\mathcal{G}_{k,c}(\tau) = \{i \in \mathcal{T}_k : a_{i,c} \le \tau\}.$$

These retained epochs define a threshold-dependent training template

$$\bar{x}^{(\tau)}_{k,c}(t) = \frac{1}{|\mathcal{G}_{k,c}(\tau)|} \sum_{i \in \mathcal{G}_{k,c}(\tau)} x_{i,c,t},$$

where candidates yielding an empty retained set are treated as inadmissible. The corresponding validation reference is computed as the pointwise median across the validation epochs,

$$\tilde{x}_{k,c}(t) = \mathrm{median}_{i \in \mathcal{V}_k}\, x_{i,c,t}.$$

The use of a median validation reference makes the criterion less sensitive to large validation-set artifacts. The validation loss for channel c is then defined as

$$\mathcal{J}_c(\tau) = \frac{1}{K} \sum_{k=1}^{K} \left[\frac{1}{T} \sum_{t=1}^{T} \left\{ \bar{x}^{(\tau)}_{k,c}(t) - \tilde{x}_{k,c}(t) \right\}^2 \right]^{1/2}.$$

The selected channel threshold is the candidate that minimizes the average validation discrepancy,

$$\tau_c^\star = \arg\min_{\tau \in \mathcal{Q}_c} \mathcal{J}_c(\tau).$$

In practice, this objective may be searched by sequential Bayesian optimization or by a random search over candidate peak-to-peak thresholds; both choices retain the same validation objective and rejection rule.

After the thresholds $\{\tau_c^\star\}_{c=1}^{C}$ have been estimated, each epoch receives a channel-level artifact indicator

$$z_{i,c} = \mathbf{1}\{a_{i,c} > \tau_c^\star\},$$

where $\mathbf{1}\{\cdot\}$ is the indicator function. The present pipeline uses the union of the channel-level decisions to form the epoch-level screening mask,

$$b_i = \mathbf{1}\left\{ \sum_{c=1}^{C} z_{i,c} \geq 1 \right\} = \mathbf{1}\left\{ \exists c \in \mathcal{C} : a_{i,c} > \tau_c^\star \right\}.$$

The set of suspicious epochs is therefore

$$\mathcal{B} = \{ i \in \mathcal{E} : b_i = 1 \}.$$

These epochs are interpreted as likely blink-contaminated or artifact-heavy segments. They provide the restricted data subset from which the next stage estimates sample-level blink-region thresholds.

3.6 Computation of Sample-Level Blink-Region Threshold Candidates

The second stage converts the epoch-level artifact decision into a sample-level thresholding problem. The autoreject-style screening in the previous stage identifies a set of suspicious epochs $\mathcal{B}$, but its thresholds $\tau_c^\star$ are defined on peak-to-peak amplitudes over whole epochs. Such thresholds are therefore appropriate for deciding whether an epoch is abnormal, but they are not directly interpretable as sample-wise blink crossing levels. Stage B estimates a separate channel-specific threshold in the amplitude domain of the time series itself.

Let $\mathcal{S}$ denote the set of epochs used for estimating the sample-level threshold. In the primary analysis, $\mathcal{S} = \mathcal{B}$, so that the estimate is driven by epochs already identified as likely to contain blink-related activity. If too few suspicious epochs are available, the method falls back to the full valid epoch set $\mathcal{E}$ to avoid estimating a threshold from an insufficient sample:

$$\mathcal{S} = \begin{cases} \mathcal{B}, & |\mathcal{B}| \geq n_{\min}, \\ \mathcal{E}, & |\mathcal{B}| < n_{\min}, \end{cases}$$

where $n_{\min}$ is the minimum number of screened epochs required for a stable estimate.

For each channel c, all samples from the selected epochs are pooled into a one-dimensional empirical sample set

$$\mathcal{Y}_c = \{x_{i,c,t} : i \in \mathcal{S},\ 1 \leq t \leq T\}.$$

This pooling step marks the methodological transition from epoch-level rejection to sample-level localization: the quantities used in Stage B are no longer peak-to-peak ranges $a_{i,c}$, but the individual signal amplitudes $x_{i,c,t}$ that will later be thresholded to form candidate blink regions.

The sample-level threshold is estimated using a robust location-plus-dispersion form. Let ξ_c be the channel-specific center of the sample distribution. The default choice is the median,

$$\xi_c = \mathrm{median}_{y \in \mathcal{Y}_c}\, y,$$

although the arithmetic mean may be substituted when a mean-centered threshold is required for comparison. Dispersion is estimated by the median absolute deviation (MAD),

$$\mathrm{MAD}_c = \mathrm{median}_{y \in \mathcal{Y}_c} \left| y - \mathrm{median}_{u \in \mathcal{Y}_c}\, u \right|,$$

and is rescaled to a standard-deviation-compatible quantity by

$$\sigma_c^{\mathrm{rob}} = 1.4826\ \mathrm{MAD}_c\,.$$

The constant 1.4826 is the usual normal-consistency factor for the MAD. For a sensitivity parameter $k > 0$, the resulting sample-level blink-region threshold candidate for channel c is

$$\theta_c = \xi_c + k\sigma_c^{\mathrm{rob}}.$$

The set

$$\Theta = \{\theta_c : c \in \mathcal{C}\}$$

constitutes the channel-wise threshold candidates passed to the blink-region extraction stage.

This construction deliberately separates the statistical role of the two thresholds in the pipeline. The quantities $\tau_c^\star$ from Stage A are epoch-rejection thresholds defined on peak-to-peak amplitudes, whereas θ_c is a sample-level amplitude threshold defined on the time-series samples themselves. Estimating θ_c from $\mathcal{S}$ concentrates the calculation on signal segments enriched for blink-like excursions, while the robust center and MAD-scaled dispersion reduce the influence of isolated extreme peaks within those same segments.

3.7 Extraction of Blink Regions

The final stage applies the sample-level thresholds $\Theta = \{\theta_c\}_{c=1}^{C}$ to obtain channel-wise blink regions. For each channel c, the epochs selected for localization are ordered as $\mathcal{S} = \{i_1, \ldots, i_R\}$, where $R = |\mathcal{S}|$. Their samples are concatenated into a one-dimensional signal

$$u_c[n] = x_{i_r,c,t}, \qquad n = (r-1)T + t, \quad r \in \{1, \ldots, R\}, \quad t \in \{1, \ldots, T\}.$$

Thus $u_c \in \mathbb{R}^{RT}$ is the channel-specific signal on which blink onsets and offsets are estimated. If the analysis is run in the fallback mode described above, $\mathcal{S}$ is the full valid epoch set; otherwise it is the set of epochs identified as suspicious in Stage A.

Candidate blink samples are defined by a threshold-crossing indicator

$$h_c[n] = \mathbf{1}\{u_c[n] > \theta_c\}, \qquad n = 1, \ldots, RT.$$

After any required polarity normalization, $h_c[n] = 1$ indicates that the sample amplitude exceeds the channel-specific blink-region threshold. Blink regions are then defined as maximal contiguous runs of supra-threshold samples. A candidate interval for channel c is written as the half-open index range $[s_{c,r}, e_{c,r})$, where

$$h_c[n] = 1 \quad \text{for all } n = s_{c,r}, \ldots, e_{c,r} - 1,$$

with either $s_{c,r} = 1$ or $h_c[s_{c,r} - 1] = 0$, and either $e_{c,r} = RT + 1$ or $h_c[e_{c,r}] = 0$. Equivalently, the onset $s_{c,r}$ is the first sample in a run that crosses above θ_c, and the offset $e_{c,r}$ is the first subsequent sample at which the signal has returned to the sub-threshold state.

To suppress brief threshold crossings that are unlikely to represent complete blinks, each candidate interval is required to exceed a minimum duration. Let f_s be the sampling frequency and let $\ell_{\min}$ be the minimum blink duration in seconds. The corresponding sample length is

$$L_{\min} = \ell_{\min} f_s.$$

The accepted blink-region set for channel c is

$$\mathcal{R}_c = \left\{(s_{c,r}, e_{c,r}) : e_{c,r} - s_{c,r} > L_{\min}\right\}.$$

This duration gate is applied independently for each channel. Consecutive supra-threshold samples are merged into the same region by construction, while two runs separated by at least one sub-threshold sample are treated as distinct candidate blink regions.

Finally, the accepted intervals are mapped from concatenated coordinates back to epoch-local coordinates using the known epoch boundaries. If an accepted onset $s_{c,r}$ lies in the concatenated segment corresponding to epoch i_q, its within-epoch onset sample is

$$\hat{s}_{c,r} = s_{c,r} - (q-1)T,$$

and its onset time is $(\hat{s}_{c,r} - 1)/f_s$ seconds relative to the beginning of that epoch. The interval duration is

$$\hat{d}_{c,r} = \frac{e_{c,r} - s_{c,r}}{f_s}.$$

The final output of this stage is therefore a channel-indexed collection of blink regions, each represented by an epoch identifier, a channel label, an onset index or time, and an offset or duration. The extraction remains channel-wise throughout: the threshold θ_c estimated for channel c is applied only to that channel's signal, and the resulting regions $\mathcal{R}_c$ are retained as channel-specific blink candidates.

3.8 Detector Configurations

Four detector configurations were compared under identical preprocessing and epoching. The proposed methods were Proposed-Med, which implemented the three-stage pipeline with a median centre, and Proposed-Mean, which used a mean centre. The conventional baselines were BLINKER-concat and MNE-annot.

3.9 Evaluation and Statistical Analysis

Detection was scored at the event level: a detected interval that overlapped an unmatched ground-truth interval was counted as a true positive, unmatched detections were counted as false positives, and unmatched ground-truth intervals were counted as false negatives. Event-level precision, recall, and F_1 were macro-averaged across sessions, and event-level F_1 was preferred over accuracy because blink events were sparse relative to non-blink samples. Sensitivity to boundary tolerance was assessed across intersection-over-union thresholds 0.0, 0.1, 0.2, 0.3, 0.5. Methods were compared using paired two-tailed Wilcoxon signed-rank tests on session-level F_1, with Bonferroni correction applied to six comparisons for the four-method pairwise tests and to four comparisons for the epoch sweep against the 30 s reference, for which the corrected alpha was 0.0125; rank-biserial effect sizes were also reported.

4 Experiments and Results

Table 1: Summary of the result-section experiment artifacts and the scripts that produce them, evaluated on Raja and Cao2018.

Artifact	Experiment and Description
`41_exp1_exp2_strategy_comparison.py`	**Main strategy comparison:** Compares the four visible conditions (BLINKER-concat, MNE-annot, Proposed-Mean, Proposed-Med) on Raja and Cao2018 using 30-second epochs. Outputs per-session results and summary macro/micro metrics in `runs/exp41_cao_30s/`.
`40_exp1_epoch_duration.py`	**Epoch-duration sensitivity:** Evaluates Proposed-Med at 30, 40, and 60 seconds on Raja and Cao2018. Two-tailed Wilcoxon tests compare each non-reference duration against the 30-second reference. Outputs are in `runs/exp40_cao/`.
`42_boundary_tolerance.py`	**Boundary-tolerance sensitivity:** Evaluates Proposed-Med over IoU thresholds 0, 0.1, 0.2, 0.3, and 0.5 using 30-second epochs on Raja and Cao2018. Outputs are in `runs/exp42_cao_30s/`.

4.1 Experimental Setup

All experiments are evaluated on the driving corpus described in Section 3.2. Unless otherwise stated, each continuous session is divided into non-overlapping 30-second epochs (selected by Experiment 1), and the evaluation is conducted at the event level using the overlap-based matching criterion introduced in Section 3.1.

Conditions. Four conditions are compared throughout the main experiments.

BLINKER-concat. The BLINKER algorithm is applied to the full concatenation of all valid epochs, treating the session as a single continuous signal and deriving a single threshold from the entire recording (Kleifges et al., 2017). This condition represents the current practice of running a continuous-signal detector on epoch-structured data without any epoch-aware adaptation.

MNE-annot. The MNE-Python `annotate_amplitude` routine, which marks contiguous segments whose peak-to-peak amplitude exceeds a configurable threshold, used here with its default parameterisation. This baseline represents widespread preprocessing practice in the neuroimaging community and provides a reference point grounded in community tooling.

Proposed-Mean. The full three-stage pipeline described in Section 3, with the sample-level threshold at Stage B estimated from the arithmetic mean and standard deviation of samples pooled from suspicious epochs.

Proposed-Med. The three-stage pipeline with the sample-level threshold at Stage B estimated from the median and median absolute deviation (MAD). This is the primary proposed configuration.

4.2 Evaluation Metrics

Event-level precision, recall, and F1. A detected interval is counted as a true positive if it overlaps at least one unmatched ground-truth blink region; unmatched detections are false positives, and unmatched ground-truth regions are false negatives. Per-session precision, recall, and F1 are computed from these event-level counts.

Macro versus micro F1. We adopt **macro-averaged F1** as the primary reporting metric. Macro F1 computes precision, recall, and F1 independently for each session and then reports the unweighted mean of the session-level F1 scores. This gives every session equal weight regardless of how many blink events it contains. The goal of epoch-based blink detection in driving studies is to produce a detector that works reliably session by session, not one that merely accumulates favourable aggregate counts. Micro F1, which pools all true positives, false positives, and false negatives across sessions before computing a single ratio, is dominated by sessions with large blink counts and can mask consistent failure modes on sessions with atypically few blinks. Both metrics are reported, but all hypothesis tests are conducted on the vector of session-level F1 scores, for which each session contributes exactly one observation.

Wilcoxon signed-rank test. Pairwise comparisons between conditions use the Wilcoxon signed-rank test on matched pairs of session-level F1 scores. The test is preferred over a paired t-test because session-level F1 distributions are bounded and often skewed. For comparisons in which the proposed method is hypothesised to outperform a baseline, a one-tailed test is used; stability analyses use two-tailed tests. Bonferroni correction is applied for the number of conditions compared within each experiment, and we report exact p-values alongside the matched-pairs rank-biserial correlation r as an effect-size estimate.

4.3 Experiment 1: Compare different Strategy

4.3.1 Strategy A: Naive Epoch Concatenation as a Proxy for Existing Practice

Motivation. In driving and other epoch-structured paradigms, it is common to run a continuous-signal blink detector on the raw or minimally preprocessed recording and subsequently map detections back to the epoch grid. BLINKER-concat embodies this practice: its threshold is derived from a signal pool that mixes blink-heavy and blink-free epochs. The central claim of the proposed pipeline is that this mixing dilutes the amplitude statistics used for threshold estimation, which may distort threshold placement and shift the precision-recall trade-off. Experiment 1 tests this claim directly.

Hypothesis. Naive concatenation will produce systematically lower session-level F1 than the proposed epoch-aware pipeline through an unfavourable precision-recall trade-off.

Design. All four conditions are applied to every session under 30-second epochs. Per-session macro F1, precision, and recall are summarised in Table 2.

4.3.2 Strategy B: MNE-Based Annotation as a Community Baseline

Motivation. MNE-Python is the most widely used open-source EEG analysis platform in neuroscience. Its `annotate_amplitude` routine provides a threshold-based artifact marking mechanism that many practitioners reach for when a specialised blink detector is unavailable. Establishing its performance under an event-level blink criterion, for which it was not designed, characterises the gap that specialised methods must close.

Design. The MNE-annot condition is drawn from Table 2. As a secondary diagnostic, the detection timestamps produced by MNE-annot are aligned to those from Proposed-Med, and the distribution of temporal offsets at onset and offset boundaries is reported. Systematic offsets indicate whether MNE-annot consistently detects the right events but with shifted boundaries, or whether the discrepancies arise from entirely different detection triggers.

4.3.3 Strategy F

Table 2: Main comparison on 30-second epochs. Values are macro-averaged across sessions. Best macro-F1 within each dataset block is shown in **bold**. Pairwise Wilcoxon tests use the visible four-condition family, $C(4,2) = 6$ comparisons; exact Bonferroni-corrected p-values are listed below.

Dataset	Condition	n	Macro precision	Macro recall	Macro F1
Raja	BLINKER-concat	46	0.668643	0.959200	0.758780
Raja	MNE-annot	46	0.695237	0.642037	0.609699
Raja	Proposed-Mean	46	0.911370	0.793596	0.817118
Raja	Proposed-Med	46	0.907756	0.813874	**0.830510**
Cao2018	BLINKER-concat	58	0.734429	0.986020	0.824156
Cao2018	MNE-annot	58	0.821379	0.887149	0.838279
Cao2018	Proposed-Mean	58	0.899163	0.883237	0.882237
Cao2018	Proposed-Med	58	0.891602	0.911316	**0.895359**
Pooled	BLINKER-concat	104	0.705331	0.974157	0.795239
Pooled	MNE-annot	104	0.765586	0.778734	0.737176
Pooled	Proposed-Mean	104	0.904562	0.843588	0.853434
Pooled	Proposed-Med	104	0.898747	0.868216	**0.866676**

Bonferroni-adjusted p-values: Raja: BLINKER-concat vs MNE-annot 0.808546603; Proposed-Mean vs BLINKER-concat 0.134535249; Proposed-Med vs BLINKER-concat 0.040217714; Proposed-Mean vs MNE-annot 0.011472361; Proposed-Med vs MNE-annot 0.002441489; Proposed-Mean vs Proposed-Med 0.000504607. Cao2018: BLINKER-concat vs MNE-annot 0.063012150; Proposed-Mean vs BLINKER-concat 0.000445011; Proposed-Med vs BLINKER-concat 0.000026958; Proposed-Mean vs MNE-annot 0.299755449; Proposed-Med vs MNE-annot 0.031506075; Proposed-Mean vs Proposed-Med 0.004596023. Pooled: BLINKER-concat vs MNE-annot 1.000000000; Proposed-Mean vs BLINKER-concat 0.000230102; Proposed-Med vs BLINKER-concat 0.000006588; Proposed-Mean vs MNE-annot 0.004704930; Proposed-Med vs MNE-annot 0.000126287; Proposed-Mean vs Proposed-Med 0.000001385.

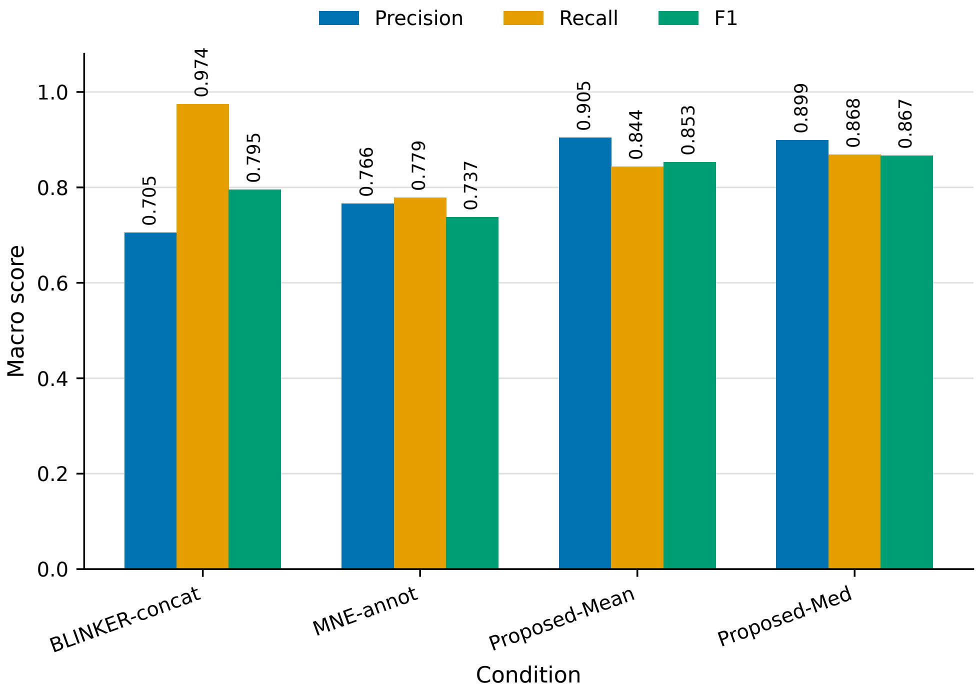

Figure 1: Macro-averaged precision, recall, and F_1 for the four conditions over all 104 sessions.

Across all 104 Raja+Cao2018 sessions, the macro-averaged results in Table 2 and Figure 1 showed a clear separation among the four evaluated conditions. Proposed-Med achieved the best macro-F_1 and, by paired Wilcoxon with Bonferroni correction over six comparisons, significantly outperformed BLINKER-concat, MNE-annot, and Proposed-Mean, obtaining precision 0.8987, recall 0.8682, and F_1 0.8667, compared with Proposed-Mean, which obtained precision 0.9046, recall 0.8436, and F_1 0.8534. Both three-stage configurations exceeded the BLINKER-concat and MNE-annot baselines in macro-F_1, with BLINKER-concat obtaining precision 0.7053, recall 0.9742, and F_1 0.7952, and MNE-annot obtaining precision 0.7656, recall 0.7787, and F_1 0.7372. BLINKER-concat reached the highest recall across conditions, but this was accompanied by the lowest precision, indicating that its high sensitivity came with a comparatively larger false-positive burden.

Wilcoxon signed-rank tests on session-level F_1, with Bonferroni correction over the six pairwise comparisons among the four conditions, were conducted as summarized in Table 2. The pooled Bonferroni-adjusted results showed significant differences for Proposed-Med versus BLINKER-concat ($p = 6.6e-6$), Proposed-Med versus MNE-annot ($p = 1.3e-4$), Proposed-Mean versus BLINKER-concat ($p = 2.3e-4$), Proposed-Mean versus MNE-annot ($p = 4.7e-3$), and Proposed-Mean versus Proposed-Med ($p = 1.4e-6$, two-sided, Proposed-Med higher), indicating that Proposed-Med was the best-performing non-baseline method and significantly outperformed BLINKER-concat, MNE-annot, and Proposed-Mean after Bonferroni correction.

4.4 Experiment 2: Effect of Threshold Estimator at Stage B

Motivation. For Strategy F, Stage B estimates the sample-level threshold from samples pooled across the suspicious epoch set $\mathcal{S}$. Because $\mathcal{S}$ is selected precisely because it contains high-amplitude excursions, the pooled distribution is not symmetric around a clean baseline: it is contaminated by the very peaks the threshold is meant to capture. The mean and standard deviation are sensitive to these extreme values; the median and MAD are not.

Hypothesis. Proposed-Med will outperform Proposed-Mean on precision—because the MAD-based threshold is less inflated by within-epoch peaks—while achieving comparable or superior recall, leading to a higher F1. The benefit of the robust estimator is expected to be larger for sessions whose suspicious-epoch pools contain extreme outlier amplitudes.

Design. Proposed-Mean and Proposed-Med are compared using the Wilcoxon signed-rank test on session-level F1 (Table 2). A supplementary scatter plot shows the per-session F1 difference against the 95th-percentile amplitude in the suspicious-epoch pool, used here as a proxy for outlier severity.

4.5 Experiment 3: Stability Across Epoch Durations

Motivation. Epoch duration is routinely chosen on paradigm-specific or theoretical grounds and is rarely optimised for the blink detector itself. In driving studies, epoch lengths ranging from 30 seconds to several minutes appear in the literature. A robust pipeline should produce stable event-level F1 across this range, because the underlying physiology of blinks does not change with the epoch grid imposed during analysis.

Design. Proposed-Med is evaluated under epoch durations of 30, 40, and 60 seconds. For each duration the full pipeline is re-run from scratch, since epoch-level peak-to-peak statistics change with epoch length. Primary outcomes are per-session macro F1; secondary outcomes are the number of suspicious epochs returned by Stage A and the estimated sample-level threshold θ_c. Table 3 reports results and two-tailed Wilcoxon p-values relative to the 30-second reference.

Table 3: Macro F1 of Proposed-Med across epoch durations. p-values (Wilcoxon, two-tailed) compare each duration against the 30-second reference. **Bold** marks the best epoch within each dataset block.

Dataset	Epoch duration	n	Macro F1	p vs. 30 s
Raja	10 s	46	0.797538	0.094837587
Raja	20 s	46	0.823601	0.851877758
Raja	30 s	46	0.830510	reference
Raja	40 s	46	**0.833218**	0.334501904
Raja	60 s	46	0.822354	0.672886398
Cao2018	10 s	58	0.884117	0.050265194
Cao2018	20 s	58	0.890888	0.801329921
Cao2018	30 s	58	**0.895359**	reference
Cao2018	40 s	58	0.877670	0.735610872
Cao2018	60 s	57	0.871752	0.759688011
Pooled	10 s	104	0.845822	0.017543417
Pooled	20 s	104	0.861126	0.837339163
Pooled	30 s	104	**0.866676**	reference
Pooled	40 s	104	0.858009	0.373667028
Pooled	60 s	103	0.849691	0.782276821

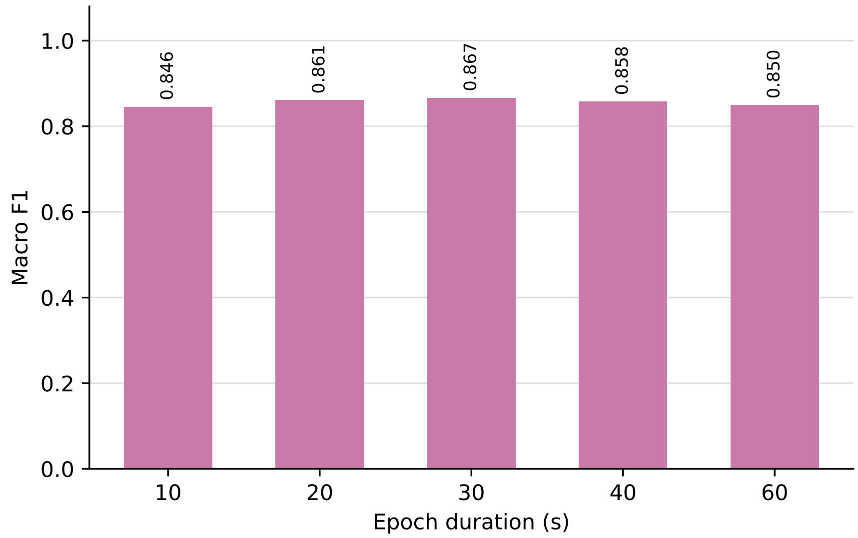

Figure 2: Macro-averaged F_1 of Proposed-Med across epoch durations (all sessions).

The effect of epoch duration was evaluated at 10, 20, 30, 40, and 60 s, as summarised in Table 3 and Figure 2. Proposed-Med pooled macro-F_1 was 0.8458, 0.8611, 0.8667, 0.8580, and 0.8497 across 10, 20, 30, 40, and 60 s, respectively. After Bonferroni correction (corrected $\alpha = 0.0125$), two-tailed Wilcoxon tests against the 30 s reference were non-significant for all non-reference durations: 10 s (p=0.018), 20 s (p=0.837), 40 s (p=0.374), and 60 s (p=0.782). Note that the shortest 10 s epochs showed a slight trend toward lower F_1 (the largest negative effect size) but did not differ significantly from 30 s after correction. Performance was therefore statistically stable across 10–60 s, with 30 s giving the best pooled macro-F_1 and selected as the primary setting.

4.6 Experiment 4: Stability Across Peak-Side Boundary Tolerance

Motivation. Event-level matching requires specifying how much temporal mismatch between a detected interval and a ground-truth interval is tolerable. Blink boundaries in annotation are inherently uncertain: different annotators and annotation tools locate onset and offset slightly differently. A detector whose F1 score is highly sensitive to the matching tolerance is fragile in practice, because its apparent performance depends on an arbitrary methodological choice rather than on true detection quality.

Design. Proposed-Med is evaluated under five event-matching strictness settings, expressed as intersection-over-union (IoU) thresholds $\{0.0, 0.1, 0.2, 0.3, 0.5\}$ for the overlap between each detected interval and a ground-truth interval. Lower IoU thresholds treat any overlap as a match; higher thresholds require a larger fraction of overlap and therefore penalise boundary mismatch more aggressively. We report macro-averaged precision, recall, and F1 at each IoU threshold, and the range of macro F1 across the five levels to quantify sensitivity to this evaluation choice.

4.7 Experiment 5: Cross-Dataset Generalisation

Motivation. A detector that is tuned and reported on a single corpus may not transfer to data collected under different acquisition conditions. Because our benchmark spans a closed corpus

Table 4: Boundary-tolerance analysis for Proposed-Med on Raja, Cao2018, and the pooled 104-session set at 30 s epochs. Macro-F_1 is shown for increasing event-matching IoU thresholds; macro-F_1 falls as the overlap criterion tightens.

Dataset	n	IoU 0.0	IoU 0.1	IoU 0.2	IoU 0.3	IoU 0.5
Raja	46	0.892100	0.830510	0.780355	0.679374	0.329248
Cao2018	58	0.963797	0.895359	0.859942	0.712317	0.210391
Pooled	104	0.932085	0.866676	0.824740	0.697746	0.262962

(Raja) and an open corpus (Cao2018), we can quantify the generalisation gap directly as the per-condition difference in macro-F1 between the two datasets.

Cross-dataset generalisation was assessed by comparing macro-F_1 between Raja and Cao2018, with the gap defined as Raja minus Cao2018 (Table 5; Figure 3). Performance was higher on Cao2018 than on Raja for every condition in this comparison. Proposed-Med generalised with a small, consistent gap, obtaining 0.8305 on Raja and 0.8954 on Cao2018, corresponding to a gap of -0.0648, while Proposed-Mean achieved 0.8171 and 0.8822 with a gap of -0.0651. BLINKER-concat produced macro-F_1 values of 0.7588 and 0.8242, giving a gap of -0.0654. Thus, the proposed methods kept a small, consistent gap across the two corpora of about 0.065, whereas MNE-annot was far less stable across datasets, with macro-F_1 values of 0.6097 on Raja and 0.8383 on Cao2018 and the largest gap of -0.2286.

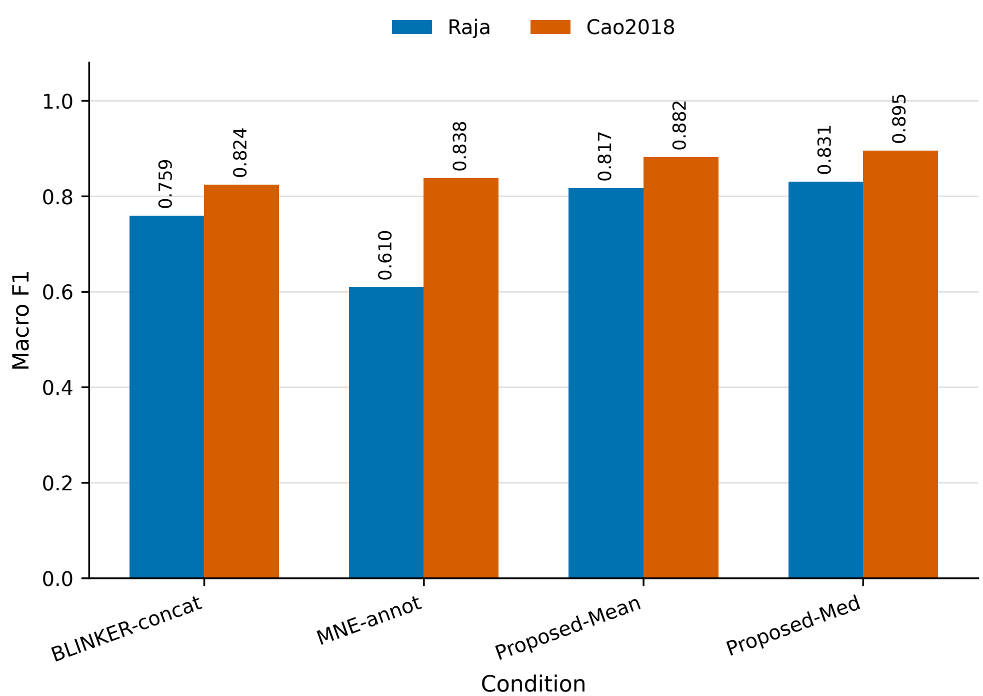

Figure 3: Macro-averaged F_1 per condition on Raja and Cao2018.

Table 5: Extra analysis: cross-dataset generalisation gap (macro-F1, 30 s epochs). Gap = Raja − Cao2018.

Condition	Raja	Cao2018	Gap
BLINKER-concat	0.758780	0.824156	-0.065376
MNE-annot	0.609699	0.838279	-0.228580
Proposed-Mean	0.817118	0.882237	-0.065119
Proposed-Med	0.830510	0.895359	-0.064849

4.8 Recall by Blink Duration

Motivation. Using Raja and Cao2018, this analysis asks whether event-level recall differs between normal-duration and long-duration blinks (Table 6).

Across 81,852 annotated normal blinks and 7,377 long blinks (89,229 total; long blinks accounting for approximately 8.3

Table 6: Extra analysis: event-level recall split by blink duration (30 s epochs). Normal = duration < 0.5 s, Long = duration ≥ 0.5 s (the long-closure threshold). Ground truth contains 81,852 normal and 7,377 long blinks across Raja and Cao2018. Best recall per column in **bold**.

Condition	Normal recall	Long recall
BLINKER-concat	**0.9848**	**0.9676**
MNE-annot	0.7761	0.7931
Proposed-Mean	0.8186	0.7961
Proposed-Med	0.8548	0.8230

4.9 Error-Structure Decomposition

Motivation. Using Raja and Cao2018, this analysis asks how each condition's event-level errors decompose into false-positive and false-negative regimes (Table 7).

Table 7 showed distinct error regimes across the four conditions based on the mean per-session FP, FN, and FP:FN ratio. BLINKER-concat was strongly false-positive-heavy, with 244.0 FP, 14.3 FN, and an FP:FN ratio of 17.1, whereas MNE-annot was false-negative-heavy, with 95.6 FP, 190.9 FN, and a ratio of 0.50. In contrast, both proposed configurations remained false-negative-heavy: Proposed-Mean produced 49.0 FP and 157.2 FN with a ratio of 0.31, while Proposed-Med produced 55.6 FP and 126.8 FN with a ratio of 0.44. Overall, the proposed configurations traded recall for far fewer false positives than BLINKER-concat, indicating a shift from an over-detection regime toward a more conservative detection profile.

4.10 Per-Session and Per-Subject Variability

Motivation. Using Raja and Cao2018, this analysis asks how Proposed-Med performance varies across individual recordings and subjects (Table 8).

Table 8 showed that Proposed-Med exhibited substantial variability across recordings and subjects. Across 104 sessions, per-session F_1 ranged from 0.0767 to 1.0000, with a median of 0.9133. The best session was Cao2018 S53/090925m ($F_1 = 1.0000$), whereas the worst session was Raja S16/S29_20190111_034326_3 ($F_1 = 0.0767$). At the subject level, across 44 subjects, Cao2018

Table 7: Error-structure decomposition by condition. Mean false positives (FP) and false negatives (FN) are reported per session.

Condition	Mean FP/session	Mean FN/session	FP:FN	Regime
BLINKER-concat	244.0	14.2	17.119	FP-heavy
MNE-annot	95.6	190.9	0.501	FN-heavy
Proposed-Mean	49.0	157.2	0.312	FN-heavy
Proposed-Med	55.6	126.8	0.438	FN-heavy

S55 had the highest mean F_1 of 0.9930 over 1 session, while Raja S16 had the lowest mean F_1 of 0.2115 over 2 sessions; the median subject mean F_1 was 0.8869.

Table 8: Best and worst Proposed-Med sessions and subject-level summary rows across 104 Raja+Cao2018 sessions.

Scope	Dataset	Unit	n	Metric	Value
Best session	cao2018	S53/090925m	1	F1	1.0000
Worst session	raja	S16/S29_20190111_034326_3	1	F1	0.0767
Median session	all	104 sessions	104	F1	0.9133
Best subject	cao2018	S55	1	Mean F1	0.9930
Worst subject	raja	S16	2	Mean F1	0.2115
Median subject	all	44 subjects	44	Mean F1	0.8869

4.11 Channel-Selection Robustness Across Methods

Motivation. Using Raja and Cao2018, this analysis asks whether the selected best channel is stable across the four visible detection methods (Table 9).

Table 9 summarised the robustness of best-channel selection across methods. Across the 104 sessions, all four visible methods selected the same channel in 31.7

Table 9: Channel-robustness ranking stability across sessions for the four visible methods. Full agreement is the fraction of sessions where all four methods select the same best channel; per-method agreement is the mean fraction of the other three methods selecting the same best channel.

Dataset	Sessions	Full agreement	Mean pairwise	BLINKER-concat	MNE-annot	Proposed-Mean	Proposed-Med
Raja	46	14/46 (0.304348)	0.572464	0.492754	0.507246	0.637681	0.652174
Cao2018	58	19/58 (0.327586)	0.591954	0.563218	0.511494	0.637931	0.655172
Pooled	104	33/104 (0.317308)	0.583333	0.532051	0.509615	0.637821	0.653846

4.12 EEG Channel Selection

Motivation. Using Raja and Cao2018, this analysis asks which EEG channels are most frequently selected across methods and datasets (Table 10; Figure 4).

Best-channel selections were pooled over the four methods for each dataset, as summarized in Table 10 and Figure 4. On Raja (184 selections), the most frequently selected channels were E9 (48.9

Table 10: Best-channel selection frequencies pooled over the four visible methods. Region rows use only labels present in `brain_region.yaml`; unmapped labels are not forced into a cross-dataset region.

Dataset	Summary	Frequencies
Raja	Channels	E9 90/184 (0.489); E22 73/184 (0.397); E3 21/184 (0.114)
Raja	Regions	frontal_minimal 184/184 (1.000); mapped 184/184 (1.000)
Cao2018	Channels	FP1 144/232 (0.621); FP2 71/232 (0.306); F7 5/232 (0.022); F8 3/232 (0.013); FC3 2/232 (0.009)
Cao2018	Regions	No mapped selections; mapped 0/232 (0.000)

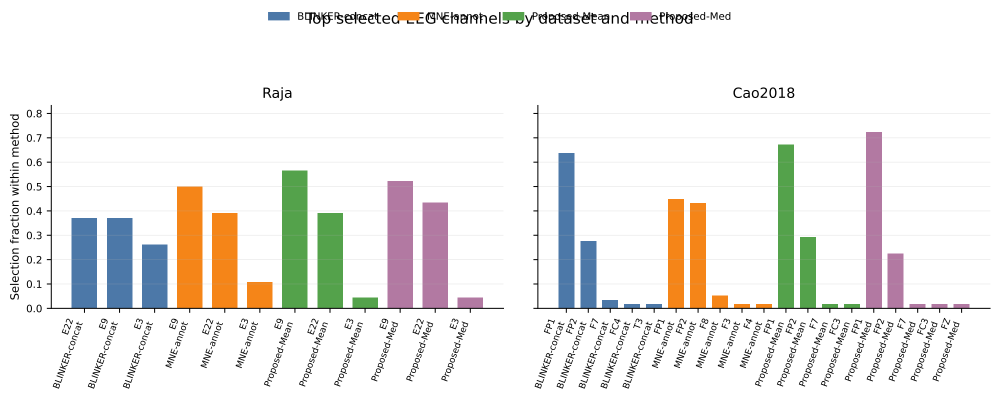

Figure 4: Channel-selection frequency by dataset and method.

5 Discussion

The main implication is that epoch-aware thresholding provides a clear advantage over naive baselines for blink detection in epoch-structured EEG data. Proposed-Med achieved a macro-F_1 of 0.8667, compared with 0.7952 for BLINKER-concat and 0.7372 for MNE-annot, indicating that adapting the decision rule to the epoch structure improved performance over applying baseline detectors without such adaptation. This pattern is consistent with the observation that running a continuous-signal detector such as BLINKER on epoch-structured data underperformed epoch-aware estimation (Kleifges et al., 2017). At the same time, Proposed-Med was the best-performing non-baseline method by F_1 and significantly outperformed BLINKER-concat, MNE-annot, and Proposed-Mean under Wilcoxon tests with Bonferroni correction.

Blink amplitude varies across subjects, electrodes, and sessions, so a single global threshold becomes sensitive to the composition of epochs used to estimate amplitude statistics (Croft and Barry, 2000; Guttmann-Flury et al., 2019). When blink-free and blink-heavy epochs are pooled, the mixture dilutes the amplitude distribution and may distort the global threshold, which can lower recall by making smaller or session-specific blinks less likely to exceed the decision boundary (Klein and Skrandies, 2013; Guttmann-Flury et al., 2019). Stage A addresses this potential bias by concentrating Stage B estimation on suspicious epochs rather than on the full clean–blink mixture, while the median/MAD estimator resists within-epoch peak contamination during threshold estimation (Klein and Skrandies, 2013). This mechanism is consistent with why Proposed-Med edges Proposed-Mean (pooled macro-F_1 0.8667 for Proposed-Med vs 0.8534 for Proposed-Mean) without requiring a stronger claim about universal superiority.

The channel-selection analysis indicated that best-channel selection concentrated on frontal and frontopolar sites in both corpora (Table 10). This pattern was most directly interpretable for

Cao2018, where all 32 channels were available and selection genuinely concentrated on the frontopolar channels FP1 (62.1

The Raja to Cao2018 comparison showed different degrees of cross-dataset gap across methods, with performance higher on Cao2018 than on Raja: the macro-F_1 gap (Raja minus Cao2018) was about -0.065 for Proposed-Med and -0.229 for MNE-annot (Z. Cao et al., 2019). This pattern indicated that the proposed method showed a small, consistent cross-dataset gap, supporting its consistency across the two evaluated datasets. Within the four-condition comparison, Proposed-Med was the best-performing non-baseline method and significantly outperformed BLINKER-concat, MNE-annot, and Proposed-Mean (Wilcoxon, Bonferroni).

The error-structure and blink-type-recall findings indicate that the proposed robust epoch-aware threshold produced a clean but conservative detection profile. In Table 7, BLINKER-concat was strongly false-positive-heavy, indicating over-detection, whereas MNE-annot and both proposed configurations were false-negative-heavy; therefore, the proposed methods traded some recall for far fewer false positives. This pattern is important because it suggests that the robust epoch-aware threshold is preferable in applications where false positives are costly, although it may miss a subset of true blink events. The blink-type analysis in Table 6 further showed that recall was high for normal blinks and slightly lower for long blinks for three of the four methods, with Proposed-Med achieving the best recall among the conservative methods for both normal blinks (0.8548) and long blinks (0.8230). Together, these results imply that Proposed-Med offers a balanced conservative alternative with reduced over-detection, while long blinks remain a small residual weakness that merits future attention.

Performance was statistically stable across 30, 40, and 60 s epochs (Wilcoxon tests against 30 s non-significant; F_1 range 0.0170), indicating that the method does not require careful epoch tuning in fatigue-driving EEG pipelines, and 30 s was selected as the primary setting (Alyan, 2023; Tian and J. Cao, 2021). Its single-channel and threshold-based design is practically attractive because it is cheap for real-time use. However, the evidence remains limited because evaluation is restricted to two driving corpora only, and long-blink recall and channel selection still require further work.

To our knowledge, this is among the first studies to explicitly account for the epoch structure of EEG recordings when estimating the amplitude threshold used to detect blink candidates. Conventional thresholding methods estimate a single global threshold from the entire recording; by contrast, our three-stage pipeline first screened suspicious epochs in Stage A and then estimated the sample-level threshold from those epochs in Stage B, directly addressing the dilution bias introduced when blink-free and blink-heavy epochs are pooled.

The present evaluation has several limitations that should temper interpretation of the results. First, the evaluation is limited to two driving-EEG corpora, Raja (closed) and Cao2018 (open) (Z. Cao et al., 2019), so generalisation to other paradigms, devices, and populations remains untested. Second, the method is single-channel and amplitude-threshold based; it does not exploit multi-channel spatial information or waveform morphology. Third, best-channel selection is only partly method-robust, with pooled mean cross-method agreement of 0.583 and full agreement among all four methods in 31.7

6 Conclusion

This paper addressed blink detection in the epoch-structured setting, a practically important but largely overlooked regime in which a continuous recording is divided into non-overlapping analysis windows before automated processing begins. We argued that the standard practice of concatenating available epochs and deriving a single amplitude threshold from the pooled signal can be problematic, because blink-free epochs may dilute the amplitude statistics that govern threshold placement. To address this, we proposed a three-stage epoch-aware pipeline: Stage A screens epochs by their peak-to-peak amplitude using an autoreject-inspired criterion, Stage B estimates a sample-level threshold from the resulting pool of suspicious epochs using the median and median absolute deviation, and Stage C extracts blink regions and maps them back to epoch-local coordinates, with a fallback that stabilises estimation when too few suspicious epochs are available. We evaluated four conditions across two naturalistic driving corpora under an event-level overlap criterion with macro-averaged F_1 and Wilcoxon signed-rank tests. The central findings are threefold. First, the proposed median configuration improved over the BLINKER-concat and MNE-annotation baselines, primarily by recovering precision without collapsing recall. Second, the robust MAD-based estimator (Proposed-Med) was preferable to its mean-based counterpart (Proposed-Mean), consistent with the expectation that the median and MAD reduce sensitivity to within-epoch peaks in the suspicious-epoch pool. Third, Proposed-Med was the best-performing non-baseline method and significantly outperformed BLINKER-concat, MNE-annot, and Proposed-Mean under Wilcoxon signed-rank tests with Bonferroni correction, while performance was relatively stable across the tested epoch durations and showed only a small Raja–Cao2018 macro-F_1 gap. We also report an honest limitation: under the selected 30-second epoch setting, the proposed median configuration did not fully resolve all threshold-placement challenges. Characterising precisely when Stage A helps and when it is overly conservative is a clear direction for future work. Further avenues include extending the evaluation to synchronized EEG–video benchmarks with high-confidence blink-region labels, and studying threshold-family selection as a first-class problem under a shared protocol rather than as a choice absorbed into a larger pipeline.

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